Instruction Formats Based on Address Count

1 Instruction Types Based on Number of Addresses

1.1 Three-Address Instruction

$$\texttt{ADD Z, X, Y} \quad \Rightarrow \quad Z := X + Y$$

Description: Explicitly specifies two source operands and one destination. Allows more powerful single instructions.

1.2 Two-Address Instruction

$$\texttt{ADD X, Y} \quad \Rightarrow \quad X := X + Y$$

Description: One operand serves as both source and destination. Saves instruction space, but can increase instruction count.

1.3 One-Address Instruction

$$\texttt{ADD X} \quad \Rightarrow \quad \mathrm{AC} := \mathrm{AC} + X$$

Description: Uses an implicit accumulator (AC). Every operation involves AC and a memory operand.

2 Expression to Evaluate

We will evaluate:

$$X := C \times C + A \times B$$

using all three formats.

3 One-Address Machine (With Accumulator)

$$\begin{aligned}
\texttt{LOAD A} &\quad \mathrm{AC} := A \\
\texttt{MULTIPLY B} &\quad \mathrm{AC} := \mathrm{AC} \times B \\
\texttt{STORE T} &\quad M(T) := \mathrm{AC} \quad \text{(temporary result)} \\
\texttt{LOAD C} &\quad \mathrm{AC} := C \\
\texttt{MULTIPLY C} &\quad \mathrm{AC} := C \times C \\
\texttt{ADD T} &\quad \mathrm{AC} := \mathrm{AC} + T \\
\texttt{STORE X} &\quad M(X) := \mathrm{AC}
\end{aligned}$$

Explanation: This method needs multiple loads and stores due to limited instruction structure and reliance on the accumulator.

4 Two-Address Machine

$$\begin{aligned}
\texttt{MOVE T, A} &\quad T := A \\
\texttt{MULTIPLY T, B} &\quad T := T \times B \\
\texttt{MOVE X, C} &\quad X := C \\
\texttt{MULTIPLY X, C} &\quad X := X \times C \\
\texttt{ADD X, T} &\quad X := X + T
\end{aligned}$$

Explanation: This format offers improved flexibility over one-address instructions but still requires intermediate variables.

5 Three-Address Machine

$$\begin{aligned}
\texttt{MULTIPLY T, A, B} &\quad T := A \times B \\
\texttt{MULTIPLY X, C, C} &\quad X := C \times C \\
\texttt{ADD X, X, T} &\quad X := X + T
\end{aligned}$$

Explanation: The most efficient and readable format. All operands are explicitly mentioned, reducing instruction count.

6 Summary Comparison Table

Feature	One-Address Machine	Two-Address Machine	Three-Address Machine
Number of Operands	1 (Accumulator)	2 (Dest = Src1 op Src2)	3 (Dest, Src1, Src2)
Requires Accumulator	Yes	No	No
Temporary Variables	Yes (e.g., T)	Yes (T, X)	Optional
Instruction Count	High	Moderate	Low
Readability	Low	Medium	High
Efficiency	Low (more memory use)	Medium	High